

## Example in MATH500

Let  $z = 2 + \sqrt{2} - (3 + 3\sqrt{2})i$ , and let  $c = 2 - 3i$ . Let  $w$  be the result when  $z$  is rotated around  $c$  by  $\frac{\pi}{4}$  counter-clockwise.

```
[asy]
unitsize(0.6 cm);
pair C, W, Z;
Z = (2 + sqrt(2), -3 - 3*sqrt(2));
C = (2,-3);
W = rotate(45,C)*(Z);
draw(Z--C--W);
dot("c", C, N);
dot("w", W, SE);
dot("z", Z, S);
label("\frac{\pi}{4}", C + (0.6,-1));
[/asy]
```

Find  $w$ .